

Conjecture 00000004116 is false:
 $\text{asdim Conf}_1(\mathbb{R}) = 1 \neq 0 = dn - 1$

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1 The conjecture

Conjecture 00000004116 (English text, verbatim): “Definition: The asymptotic dimension is the large-scale geometric dimension invariant. Conjecture: The asymptotic dimension of $\text{Conf}_n(\mathbb{R}^d)$ is exactly $dn - 1$, coinciding with the coarse dimension, and the fractal correction term is always zero.” The Chinese text says the same.

Reading. $\text{Conf}_n(\mathbb{R}^d)$ is the ordered configuration space $\{(x_1, \dots, x_n) \in (\mathbb{R}^d)^n : x_i \neq x_j \text{ for } i \neq j\}$ with the metric induced from $(\mathbb{R}^d)^n = \mathbb{R}^{dn}$ (Euclidean). The formula is asserted for all $n, d \geq 1$. We refute its first clause, $\text{asdim Conf}_n(\mathbb{R}^d) = dn - 1$, at $n = d = 1$, so the conjunction fails. For $n = 1$ the ordered and unordered configuration spaces coincide, so the refutation does not depend on that choice either. The terms “coarse dimension” and “fractal correction term” are not defined in the text, and we do not address them.

Not refuted in Lean: the formula restricted to $n \geq 2$ or to $d \geq 2$. (The remark at the end of Section 3 explains, in prose only, why it fails there too.) If the intended range is $n \geq 2$, our formal refutation does not apply.

2 Definitions

Definition 1 (Wikipedia, “Asymptotic dimension”, formal definition). Let X be a metric space and $n \geq 0$ an integer. $\text{asdim } X \leq n$ if for every $R \geq 1$ there is a uniformly bounded cover $\mathcal{U}$ of X (that is, $\sup_{U \in \mathcal{U}} \text{diam } U < \infty$) such that every closed R -ball of X meets at most $n + 1$ members of $\mathcal{U}$. Then $\text{asdim } X$ is the least such n , and $\text{asdim } X = \infty$ if there is none.

In Lean (`AsdimLE`, `asdim`), “uniformly bounded” is $\exists D, \forall U \in \mathcal{U}, \forall x, y \in U, d(x, y) \leq D$. “Meets at most $n + 1$ members” is $\#\{U \in \mathcal{U} : U \cap \bar{B}(x, R) \neq \emptyset\} \leq n + 1$, where $\#$ is the cardinality in $\mathbb{N} \cup \{\infty\}$ (`Set.encard`). `asdim` X is the infimum in $\mathbb{N} \cup \{\infty\}$ of all such n , which is ∞ when there are none. $\text{Conf}_n(\mathbb{R}^d)$ is `Conf n d`, the subtype of injective elements of `PiLp 2 (fun _ : Fin n => EuclideanSpace R (Fin d))`, with the subspace metric.

3 Proof

Lemma 2 (`not_asdimLE_zero_of_chain`, chain argument). *Suppose that $f : \mathbb{N} \rightarrow X$ satisfies $d(f_k, f_{k+1}) \leq 1$ for all k , and that $d(f_0, f_k)$ is unbounded. Then $\text{asdim } X \leq 0$ fails.*

Proof. Take the cover $\mathcal{U}$ given for $R = 1$, with diameter bound D , and let $U \ni f_0$. By induction, $f_k \in U$ for every k . Indeed, if $f_k \in U$ and $f_{k+1} \in V \in \mathcal{U}$, then $\bar{B}(f_k, 1)$ meets both U and V , so $U = V$ by multiplicity ≤ 1 . Hence $d(f_0, f_k) \leq D$ for all k , which is a contradiction. $\square$

Lemma 3 (`asdimLE_one_of_isometry_real`). *If $\varphi : X \rightarrow \mathbb{R}$ satisfies $d(x, y) = |\varphi x - \varphi y|$, then $\text{asdim } X \leq 1$.*

Proof. For $R \geq 1$ let $L = 3R$ and $U_j = \{x : jL \leq \varphi x < (j+1)L\}$ for $j \in \mathbb{Z}$. These sets cover X and have diameter $\leq L$. Let $m = \lfloor (\varphi x - R)/L \rfloor$. If U_j meets $\bar{B}(x, R)$, then $m < j+1$ and $jL \leq \varphi x + R < (m+1)L + 2R < (m+2)L$, so $j \in \{m, m+1\}$. $\square$

Theorem 4 (`not_asdimLE_zero_conf`, `one_le_asdim_conf`). *For all $n, d \geq 1$: $\text{asdim } \text{Conf}_n(\mathbb{R}^d) \geq 1$.*

Proof. Let $e = e_1 \in \mathbb{R}^d$ and $c = (0 \cdot e, 1 \cdot e, \dots, (n-1)e)$, which is injective. Let $w = (e, \dots, e) \neq 0$. Then $f_k = c + (k/\|w\|)w$ is an injective configuration and $d(f_a, f_b) = |a - b|$. Apply Lemma 2. $\square$

Theorem 5 (`asdim_conf_one_one`). $\text{asdim } \text{Conf}_1(\mathbb{R}) = 1$.

Proof. The unique coordinate $x \mapsto x_1 \in \mathbb{R}$ is an isometry (`conf_one_one_dist`), so Lemma 3 gives ≤ 1 . The previous theorem gives ≥ 1 . $\square$

Theorem 6 (`conjecture_4116_false`). *It is not true that $\text{asdim } \text{Conf}_n(\mathbb{R}^d) = dn - 1$ for all $n, d \geq 1$: at $n = d = 1$ the value is 1, whereas $dn - 1 = 0$.*

Remark (prose only, not formalized). Wikipedia lists $\text{asdim}(\mathbb{R}^n) = n$ and states that coarsely equivalent spaces have equal asymptotic dimension. $\text{Conf}_n(\mathbb{R}^d)$ is dense in $\mathbb{R}^{dn}$, so the inclusion is a coarse equivalence, and therefore $\text{asdim } \text{Conf}_n(\mathbb{R}^d) = dn \neq dn - 1$ for every $n, d \geq 1$. The Lean proof covers only the instance $n = d = 1$ and the bound $\text{asdim} \geq 1$ for all $n, d \geq 1$.

4 Lean formalization

File `lean/Conjecture4116/Basic.lean` (208 lines, import `Mathlib` only; Lean v4.33.1, `Mathlib` v4.33.1). Main theorems: `C4116.conjecture_4116_false`, `C4116.asdim_conf_one_one`, `C4116.one_le_asdim_conf`; supporting: `not_asdimLE_zero_of_chain`, `asdimLE_one_of_isometry_real`, `not_asdimLE_zero_conf`, `conf_one_one_dist`. `lake build` succeeds, and `#print axioms` reports exactly `[propext, Classical.choice, Quot.sound]` for all three main theorems. There is no `sorry`, no `native_decide` and no added axiom.

5 Relation to the earlier submission

PR #294 (by orionsheep) made the same off-by-one observation ($\text{Conf}_1(\mathbb{R}) = \mathbb{R}$ has $\text{asdim } 1$). It was not merged because its main theorem was the numeral statement $(1 \cdot 1 - 1 = 0 \wedge 1 \neq 0) \wedge \dots$, with “no $\text{Conf}_n(\mathbb{R}^d)$, no asymptotic or coarse dimension”. The off-by-one refutation was “asserted only in prose”. This submission defines $\text{Conf}_n(\mathbb{R}^d)$ with its metric and Gromov’s asymptotic dimension in Lean. It proves $\text{asdim } \text{Conf}_1(\mathbb{R}) = 1$ (both bounds) and $\text{asdim} \geq 1$ for all $n, d \geq 1$, and it derives the refutation from these theorems.

Sources

- Wikipedia, “Asymptotic dimension”, https://en.wikipedia.org/wiki/Asymptotic_dimension (formal definition; examples “ $\text{asdim}(\mathbb{R}^n) = n$ ”; property “If X and Y are coarsely equivalent metric spaces (e.g. quasi-isometric metric spaces), then $\text{asdim } X = \text{asdim } Y$ ”).
- PR #294: <https://github.com/The-Last-Math-Competition/The-Last-Math-Competition/pull/294>.